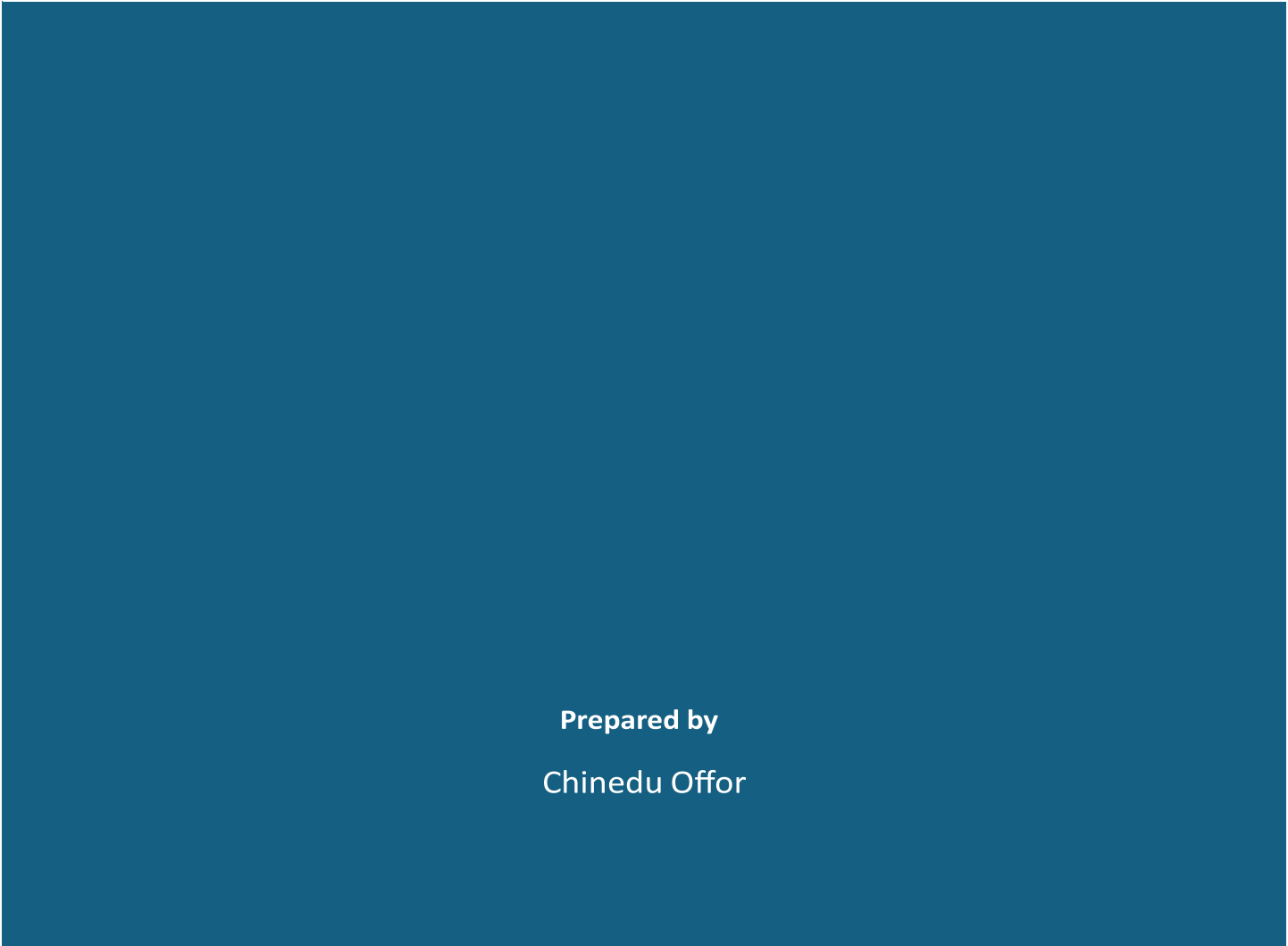




DATA ANALYTICS

Project: PERSONAL FINANCE DASHBOARD

October – December 2022 Analysis



Prepared by

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EXECUTIVE SUMMARY

The purpose of this report is to present the findings from an analysis of personal financial transactions recorded between October and December 2022, capturing both income and everyday spending. Using Microsoft Excel, the dataset was cleaned, analyzed, and transformed into an interactive dashboard that provides meaningful insights into income, expenses, savings, and spending behavior.

The analysis focused on identifying financial trends, comparing income against expenses, evaluating monthly savings performance, examining payment methods, and understanding spending patterns across different expense categories. Pivot Tables, Pivot Charts, KPI Cards, Slicers, and dashboard design techniques were used to transform raw transaction data into actionable business information.

The findings revealed that total income of **₦1,445,900** exceeded total expenses of **₦862,986**, resulting in a net savings of **₦582,914** balance throughout the reporting period. Housing & Utilities represented the highest spending category of **₦265,370**, while discretionary expenses accounted for the largest proportion of overall expenditure by **64%**. Additionally, monthly expenses increased in December **₦343,323** suggesting seasonal spending associated with year-end activities.

Overall, the dashboard demonstrates how data visualization can simplify financial analysis, support informed decision-making, and highlight opportunities to improve budgeting and long-term financial planning.

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INTRODUCTION

Financial data plays an important role in understanding spending behavior and supporting effective decision-making. Individuals and organizations rely on financial analysis to monitor income, control expenses, improve savings, and identify areas where resources can be managed more efficiently.

This project analyzes personal financial transactions collected over a three-month period, from October to December 2022. In columns A-G, the objective was to organize the data, perform meaningful analysis, and develop an interactive Excel dashboard capable of communicating financial performance in a clear and visually appealing manner.

Microsoft Excel was used throughout the project to clean the data, create Pivot Tables and Pivot Charts, calculate Key Performance Indicators (KPIs), and build an interactive dashboard with slicers for dynamic reporting. The dashboard enables users to explore financial information by month, payment method, category, and fund class while providing concise business insights to support better financial decisions.

The report documents the analytical process, presents the findings for each business question, and concludes with practical recommendations based on the results obtained from the analysis.

PROJECT OBJECTIVES

The objectives of this project were to:

- Show total expenses by month
- Show total expenses by category
- Show the count of expenses by category
- Compare total expenses against total income
- Monthly savings rate by percentage
- Spending split between credit card and cash
- Five single largest transactions size per category
- Average transaction size per category
- Applying pareto principle what share of transaction accounts for roughly 80%
- In the transaction record which what share of total expenses is fixed vs discretionary

METHODOLOGY

The analysis was completed using Microsoft Excel through the following stages:

Data Preparation

- Reviewed the dataset for duplicates, completeness and consistency.
- Organized transaction records into a structured format suitable for analysis.

Data Analysis

- Created Pivot Tables to summarize financial information.
- Developed Pivot Charts to visualize trends and comparisons.
- Calculated Key Performance Indicators (KPIs), including Total Income, Total Expenses, Net Savings, and Savings Rate.

Dashboard Development

- Designed an interactive dashboard using Pivot Charts, KPI Cards, and Slicers.
- Enabled dynamic filtering by Month, Category, Fund Class, and Payment Method.
- Included concise business insights to enhance interpretation of the results.

Reporting

- Interpreted findings from each business question.
- Developed actionable recommendations based on the analysis.

Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Dashboard Design
- PowerPoint

QUESTION 1

What are the total expenses by month?

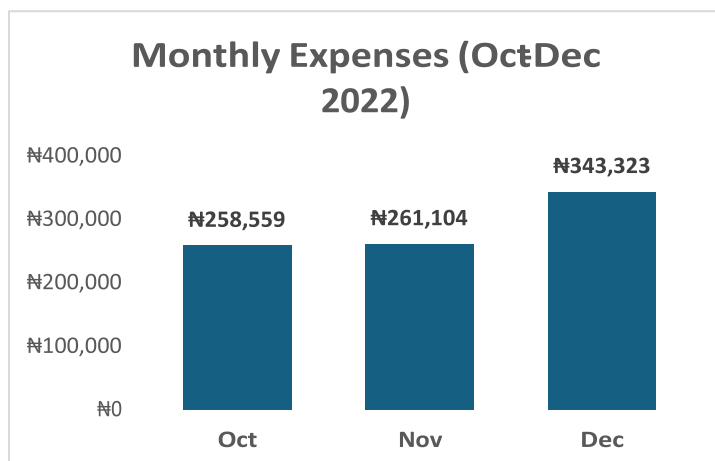


Chart Used: **Clustered Column Chart**

Why this chart?

A Clustered Column Chart was selected because it clearly compares total expenses across different months, making changes in spending easy to identify.

Analysis

Monthly expenses remained relatively stable during October and November before increasing noticeably in December.

Business Insight

The increase in December suggests higher year-end spending, likely due to seasonal activities and additional financial commitments.

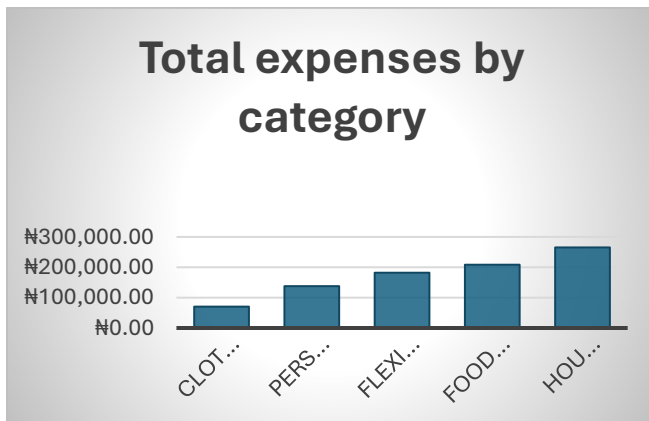
Recommendation

Develop a monthly budget that anticipates seasonal spending to avoid unnecessary financial pressure during high-expense periods.

QUESTION 2

Business Question

What are the total expenses by category?



Clustered Column Chart

Why this chart?

A Clustered Column chart was selected because it makes it easy to compare total expense across the five expense categories and quickly identify the categories with the highest spending.

Analysis

Total expenses for the period amounted to ₦862,986. Housing & Utilities recorded the highest expenditure at ₦265,370, followed by Food & Drinks at ₦207,910 and flexing at ₦182,094.

Business Insight

Housing & Utilities represents the largest expense category, making it the most significant contributor to overall spending during the reporting period.

Recommendation

Review recurring household expenses regularly to identify opportunities for cost savings without compromising essential needs.

QUESTION 3

Business Question

Which expense categories have the highest number of transactions?

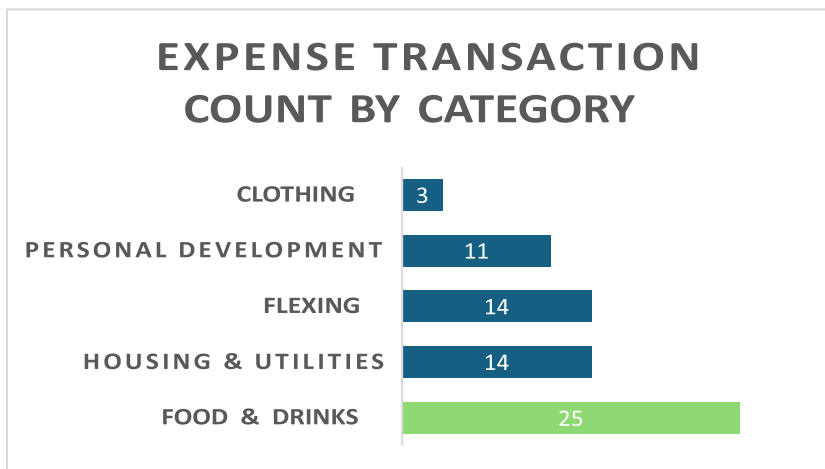


Chart Used

Bar Chart

Why this chart?

A Bar Chart was chosen because it clearly compares the frequency of transactions across different expense categories.

Analysis

Food & Drinks recorded the highest number of transactions with 25 counts, while categories such as Personal Development and Clothing had fewer transactions.

Business Insight

Although Food & Drinks may not always have the highest total spending, frequent purchases indicate an area where small daily expenses can accumulate significantly over time.

Recommendation

Track frequent low-value purchases closely and establish spending limits for categories with high transaction frequency.

QUESTION 4

Business Question

How does total income compare with total expenses?

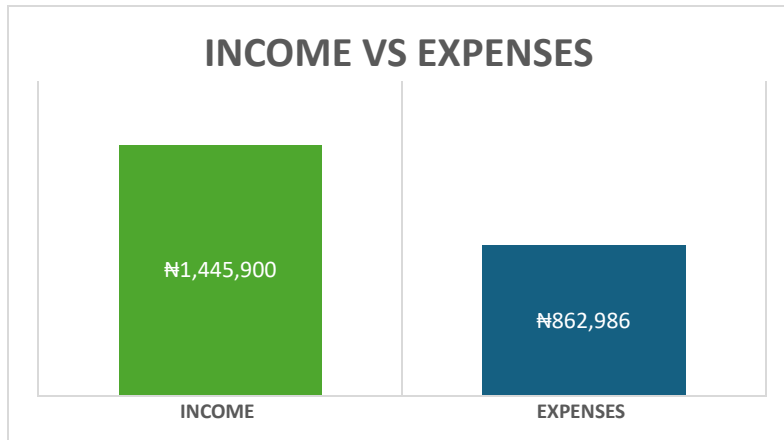


Chart Used

Stacked Column Chart

Why this chart?

A Stacked Column Chart was selected because it provides a clear comparison between total income and total expenses, making the difference between earnings and spending easy to identify.

Analysis

Total income amounted to ₦1,445,900, while total expenses were ₦862,986, resulting in net savings of ₦582,914.

Business Insight

Income exceeded expenses during the reporting period indicating that the individual maintained a positive savings position.

Recommendation

Maintain disciplined spending habits and allocate part of the surplus towards long-term savings or investment opportunities.

QUESTION 5

Business Question

What is the monthly savings rate (Income – Expenses as a percentage of Income), and is it improving or declining across the quarter?

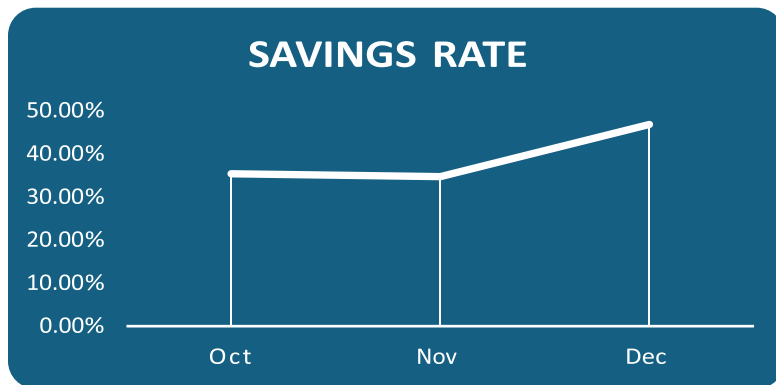


Chart Used

Line Chart

Why this chart?

A Line Chart was selected because it is the most effective visualization for displaying changes and trends over time.

Analysis

The savings rate fluctuated during the reporting period but showed an overall improvement towards the end of the quarter.

Business Insight

The improving savings rate reflects effective financial management and stronger control over expenses relative to income.

Recommendation

Continue monitoring the savings rate monthly and establish a target savings percentage to strengthen long-term financial stability.

QUESTION 6

Business Question

How do Credit Card and Cash payment methods compare?

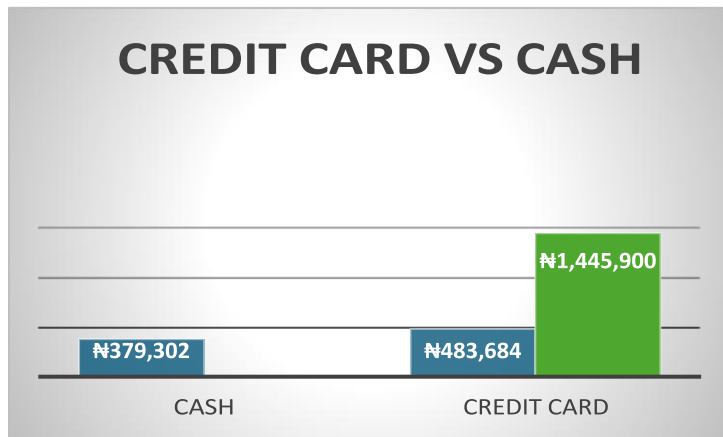


Chart Used

Clustered Column Chart Why

this chart?

A Clustered Column Chart was selected because it allows direct comparison between the two payment methods across the dataset.

Analysis

Credit Card transactions accounted for a larger proportion of spending than Cash transactions.

Business Insight

The higher use of Credit Cards suggests greater reliance on electronic payments for daily financial transactions.

Recommendation

Regularly monitor Credit Card spending to avoid unnecessary debt and ensure timely repayments.

QUESTION 7

Business Question

What are the Top 5 highest-value transactions?

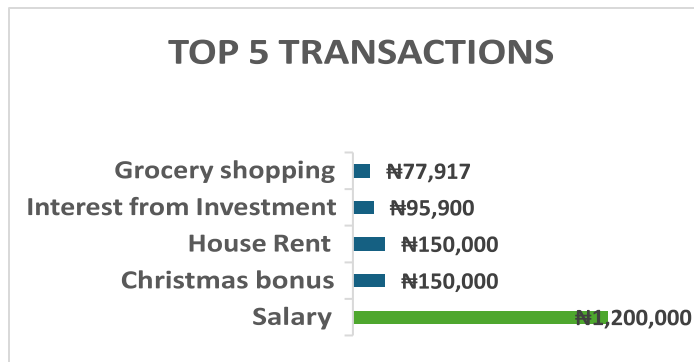


Chart Used

Bar Chart

Why this chart?

A Bar Chart was selected because it effectively ranks transactions from highest to lowest, making the largest transactions easy to identify.

Analysis

Salary was the highest-value transaction, followed by House Rent and other major financial transactions.

Business Insight

A small number of high-value transactions contribute significantly to overall cash flow.

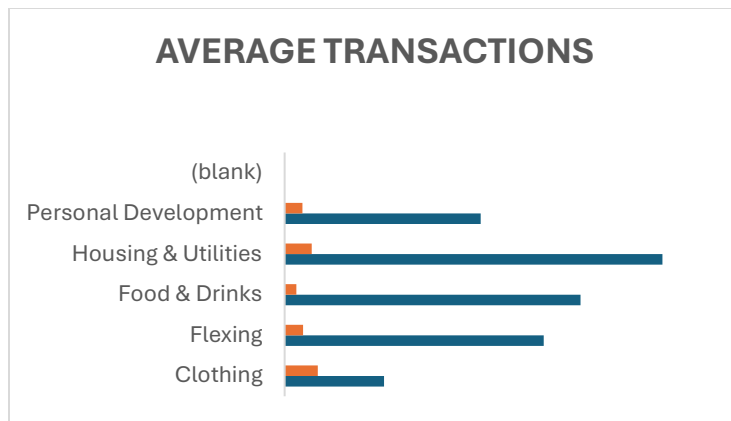
Recommendation

Pay close attention to major transactions, as they have the greatest influence on financial performance and budgeting decisions.

QUESTION 8

Business Question

Which categories have the highest average transaction value?



Bar Chart

Why this chart?

A Bar Chart was selected because it provides a clear comparison of average transaction values across expense categories.

Analysis

Housing and Utilities recorded the highest transaction followed by Food and drinks recorded the higher average transaction value, while smaller categories had significantly lower average amounts.

Business Insight

Essential expenses tend to involve larger individual payments compared to discretionary purchases.

Recommendation

Review high-value categories periodically to identify opportunities for reducing costs while maintaining essential spending.

QUESTION 9

Business Question

Does the Pareto Principle (80/20 Rule) apply to expenses?

Chart Used

Pivot Table (No Chart)

Why no chart?

A chart was not necessary because the objective was to determine how the expense categories contributed to total expenditure and whether spending followed an approximate 80/20 pattern.

The Pivot Table provided the expense totals and cumulative percentages needed to evaluate the relationship directly.

Row Labels	Sum of Amount	Running Total	Cumulative Percentage
Housing & Utilities	₦265,370.00	₦265,370.00	30.75%
Food & Drinks	₦207,910.00	₦473,280.00	54.84%
Flexing	₦182,094.00	₦655,374.00	75.94%
Personal Development	₦137,712.00	₦793,086.00	91.90%
Clothing	₦69,900.00	₦862,986.00	100.00%
Grand Total	₦862,986.00		

Analysis

Housing & Utilities, Food & Drinks, and Flexing were the three largest expense categories, accounting for approximately **75.94%** of total expenses. Adding Personal Development increased the cumulative share to **91.90%**. This indicates that spending was concentrated in a small number of categories; however, the dataset does not show a strict 80/20 relationship.

Business Insight

A relatively small number of expense categories account for most of the expenditure, meaning that monitoring the highest-spending categories can provide greater visibility into overall spending patterns.

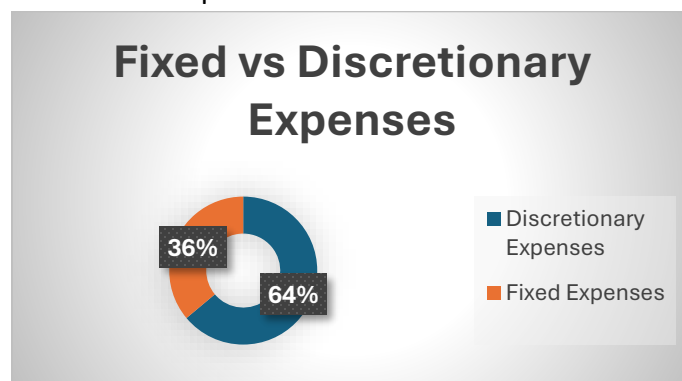
Recommendation

Focus spending reviews on the highest-cost categories, particularly Housing & Utilities, Food & Drinks, and Flexing, while continuing to monitor the remaining categories.

QUESTION 10

Business Question

How is total expenditure distributed between Fixed and Discretionary expenses?



Doughnut.

Why this chart?

A Doughnut Chart was selected because there are only two expense types, making it easy to see how total expenditure is divided between Fixed and Discretionary expenses.

Analysis

Discretionary expenses accounted for 64% of total expenditure, while Fixed expenses accounted for 36%. Discretionary spending therefore represented the larger share of overall expenses.

Business Insight

The larger share of discretionary spending indicates that a significant portion of total expenditure may be flexible and could potentially be reduced through stronger spending controls.

Recommendation

Set spending limits for discretionary purchases and regularly review non-essential expenses to identify opportunities to increase savings.

Overall Findings

- . Total Income of ₦1,445,900 exceeded Total Expenses of ₦862,986, resulting in net savings of ₦582,914.
- . Housing & Utilities recorded the highest total expenditure at ₦265,370.
- . Food & Drinks had the highest transaction frequency, with 25 transactions.
- . Salary was the largest single transaction at ₦1,200,000.
- . Monthly expenses increased significantly in December, reaching ₦343,323.
- . The monthly savings rate fluctuated during the quarter but showed an overall improvement towards the end of the reporting period.
- . Credit Card transactions accounted for a larger share of spending than Cash transactions.
- . Housing & Utilities, Food & Drinks, and Flexing accounted for approximately 75.94% of total expenses, showing that spending was concentrated in a few categories; however, the data did not demonstrate a strict 80/20 Pareto relationship.
- . Discretionary expenses represented the largest share of total expenditure at 64%, compared with 36% for Fixed expenses.

OVERALL CONCLUSION

The analysis demonstrated the value of using Microsoft Excel as a business intelligence and reporting tool for financial analysis. Through the application of Pivot Tables, Pivot Charts, KPI

calculations, and interactive dashboard design, raw financial data was transformed into meaningful information that supports informed decision-making.

The results showed that income consistently exceeded expenses during the reporting period, resulting in a healthy savings position. Housing & Utilities emerged as the largest expense category, while Discretionary spending accounted for the greatest proportion of total expenditure, representing 64% of expenses. Seasonal spending patterns were also observed, with expenses increasing during December.

The interactive dashboard provides users with a simple and effective way to monitor financial performance and identify spending trends. The insights generated from this analysis can support improved budgeting, stronger savings habits, and more strategic financial planning.
